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Highly Reliable Liquid Metal–Solid Metal Contacts with a Corrugated Single-Walled Carbon Nanotube Diffusion Barrier for Stretchable Electronics

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Liquid metals (LMs) are a special case of metals that exist in a liquid at room temperature, making them one of the most attractive conductive materials in stretchable electronics. In many cases, however, the LM attacks other metals in contact with the LM through penetration, embrittlement, and alloying. To address these critical issues, there have been efforts to introduce robust barriers, which can preserve the underlying metals without degradation. For example, graphene is employed as a flexible barrier owing to its chemical inertness and impermeability. Nevertheless, this material is difficult to utilize in stretchable electronics because its defects result in inevitable fracture, even at a low strain (< 6%). In addition, it is a challenge to pattern the graphene layer on the point-of-interest area in a facile manner. Herein, it is shown that the insertion of single-walled carbon nanotubes (SWCNTs) at the liquid metal–solid metal interface provides a proper barrier for LM under large deformation conditions. A tangled 1D structure of the SWCNTs formed from a solution process greatly suppresses the crack generation/propagation and maintains conductivity even under large strains, which facilitates the use of SWCNTs as stretchable barriers with long-term reliability through the introduction of a corrugated structure.

1. Introduction

Liquid metals (LMs) are one of the few conductive materials that have high electrical conductivity and maintain their conductivity under severe mechanical deformation. Some pure metals, such as mercury, gallium, and their alloys, have relatively low melting temperatures below room temperature, enabling them to exist as atypical liquid-phase metals. This property is useful in a variety of electrical applications where the circuit needs to be able to deform to a certain degree, namely, stretchable electronics, including stretchable antennas,[1] stretchable conductors,[2] strain sensors,[3] and pressure sensors for novel interfaces.[4] Many other materials with similar properties, such as conductive composites[5] and ionic hydrogels,[6,7] have been developed and studied as alternatives to LMs. However, there are no other materials that simultaneously possess both the metal-like high conductivity and unique mechanical properties of LMs. Because of these nonsubstitutable characteristics, LMs are used as electrical conductors in soft electronics that require high conductivity and reliability under harsh deformation conditions.

One of the main technically serious issues involving LMs in practical application is liquid metal penetration (LMP) into other solid-phase metals. When an LM is in contact with many metals, the LM [especially gallium (Ga)] easily penetrates into the other metals along the grain boundaries.[8–10] When the LM spreads, the impregnated metal film may undergo significant deformation, such as swelling, delamination,[11] or even cracking.[9] In addition, the LM can react with the surrounding metals by physical and chemical reactions, such as the dissolution of the host metal and the formation of intermetallic compounds (IMCs) and metal oxides, which can change their electrical and mechanical properties.[12–15] However, there has been a lack of discussion regarding LM-induced electrical failure in the previous studies of stretchable electronics. Several studies have implemented liquid metal–solid metal contacts (LSCs) to take advantage of the high stretchability of LMs,[16–20] but the effect of the LM on the solid metals and their long-term stability have not been considered. In the field of stretchable electronics, however, the effects of LMs should be more carefully considered than in the field of rigid electronics due to the higher level of mechanical stresses applied on devices, the preference of using thin films to bulk materials, and the relatively high subtlety of the flexible and stretchable thin films. Furthermore, since these phenomena seem to affect diverse types of LSCs (Figure S1,

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DOI: 10.1002/adfm.201806014

Supporting Information), countermeasures are indispensable for the use of LMs in stretchable electronics.

To prevent the damage of LSCs, which allows for usage of LMs in practical applications, the diffusion of the LM must be suppressed. Since the LM typically moves by wetting along the grain boundaries and reacts with other metals to form IMCs, the formation of a chemically inert and conductive diffusion barrier is one of the most promising solutions for a given LSC. In addition, to use this diffusion barrier in stretchable electronic applications, the barrier must also preserve its electrical and mechanical properties under harsh deformation. Carbon allotropes with sp^2 bonds are good candidates for stretchable diffusion barriers because of their chemical inertness, excellent mechanical properties, and high electrical conductivity. Graphene-based diffusion barriers to prevent the reaction between Galinstan (GaInSn)/aluminum and eutectic gallium–indium (EGaIn)/silver (Ag) have been investigated as promising candidates for blocking LMP.[21,22] The graphene greatly inhibits the damaging effect of the LMs at the LSC owing to its impermeable characteristics and the chemical inertness of the strong carbon sp^2 bonds without dangling bonds. However, there are still process-related limitations in producing high-quality graphene by a low-cost solution process at low temperature and ensuring a sufficient barrier thickness of the graphene layer by a transfer process. More seriously, the graphene layers have mechanical limitations such as fracture formation at very low strains (< 6%)[23] and flaws generated during both the transfer and synthesis processes that may allow the opportunity for LMs to penetrate.[24] These disadvantages hinder the direct use of pure graphene as a barrier and the usage of graphene in an ink or a composite form as an alternative method. Furthermore, these drawbacks also attenuate the benefits of graphene with regard to electrical conductivity. Consequently, the graphene barrier is very challenging for use in practical stretchable electronic applications.

Here, we report a stretchable diffusion barrier for long-term reliable LSCs and a highly stable LM-based chip bonding technique based on a corrugated single-walled carbon nanotube (SWCNT) thin film. A SWCNT diffusion barrier is formed on a corrugated silver nanoparticle (AgNP)-based thin film, which has been widely used as stretchable electrodes and contact pads.[25–27] This barrier guarantees long-term reliable LSCs, even under severe deformation conditions. EGaIn was selected as the LM for the LSC since it is highly reactive to Ag and therefore suitable for comparative analysis. Due to the chemical inertness of carbon-based materials to LMs, it is possible to form a SWCNT-based diffusion barrier for LMs, as well as the graphene barrier in the previous studies. However, unlike graphene, a SWCNT, which has a distinctive 1D elongated structure, forms a thin film in which many tubes are entangled with each other. This structure suppresses crack generation and propagation in the thin film and maintains conductivity against mechanical deformation by the crack-bridging effect.[28] Owing to the mechanical superiority of the SWCNT, it can be conformably wrinkled together on a corrugated surface of metallic thin film where the corrugation structure has a very small bending radius. Furthermore, in terms of process simplicity, the SWCNT layer is easily formed and patterned through solution processes such as spray coating or inkjet printing, conveniently securing a sufficient barrier thickness for high barrier performance.

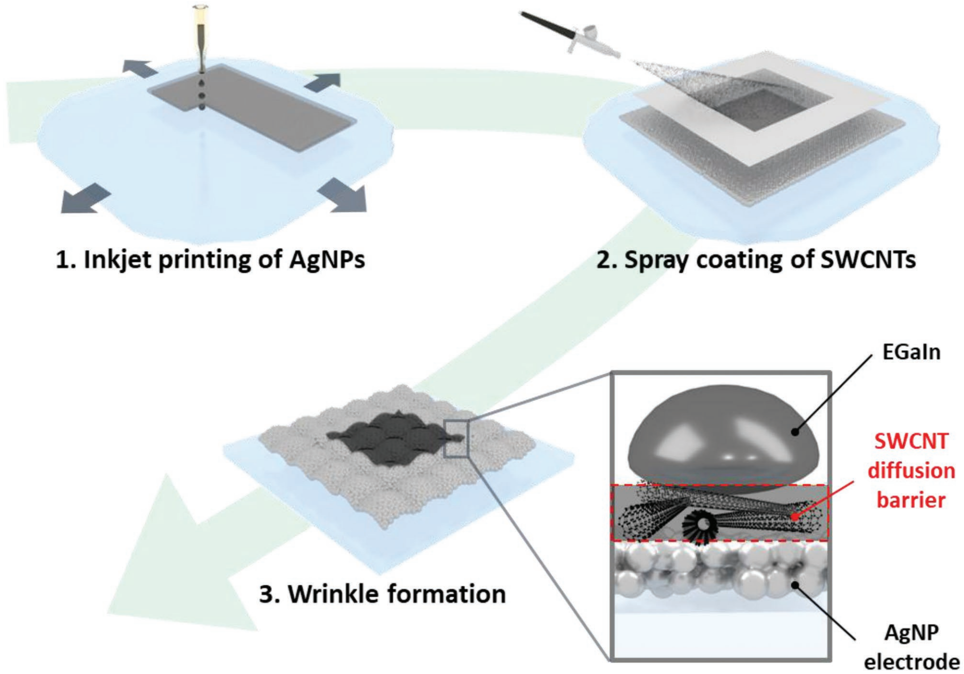
We performed a series of experiments to verify the performance of our LSC with an SWCNT barrier. Scanning electron microscopy (SEM)/optical microscopy observations and spectroscopic analyses were employed to confirm the structural properties and the barrier performance of the proposed LSC. By measuring the electrical properties of the LSCs under long-term electrical/mechanical stress, we tried to prove that the additional SWCNT layer improves the long-term stability of the LSCs. Finally, as a demonstration of a practical application, we applied the diffusion barrier to LM-based soldering technology that we developed for stretchable hybrid electronics (SHE). In the electrical bonding between a rigid chip and a soft substrate and electrodes, the LM solder could electrically connect the chip pad and soft electrodes while mechanically decoupling the rigid chips and soft substrates. As a further demonstration of LM soldering, a stretchable seven-segment circuit with LM soldering and a corrugated SWCNT diffusion barrier was successfully implemented, ultimately confirming the potential applicability and process extensibility of our method.

2. Results and Discussion

2.1. Concept and Morphology of the Corrugated SWCNT/AgNP Bilayer

Scheme 1 shows schematic diagrams of the fabrication process and the proposed structure of the corrugated SWCNT/AgNP bilayer. AgNP ink is suitable as a conductive ink in a printing process because the small particle size (≤50 nm dia.) can overcome the nozzle clogging problem.[29] With this advantage, AgNPs can be easily patterned through an inkjet printing process on a pre-stretched poly(dimethylsiloxane) (PDMS) substrate. After thermal sintering of the AgNP film, an SWCNT aqueous suspension was sprayed onto the film through spray coating. In preparation of the suspension, SWCNTs with carboxyl groups and a surfactant (sodium dodecylbenzenesulfonate, SDBS) were mixed in deionized (DI) water, and sufficient tip sonication was carried out. After the bilayer was formed, the strain was removed from the pre-stretched substrate to form corrugated structures on the SWCNT/AgNP bilayer (the detailed methodology is explained in the Experimental Section).

Figure 1 shows the morphology of the SWCNT/AgNP bilayers observed through optical and scanning electron microscopy (SEM). Because CNTs are 1D materials with a very large aspect ratio and excellent mechanical and electrical properties, facile fabrication of a well-percolated system of SWCNT layers or SWCNT composites with high conductivity is possible.[30] In particular, the nanoscale diameter of CNTs allows a thin layer to be fabricated at electrical contacts (Figure 1a,b), making it possible to both minimize any additional contact resistance caused by the barrier insertion and form a conformable intact structure on the corrugated Ag surface formed by a pre-stretching process (Figure 1c–e).[31] Moreover, due to their high surface tension,[32] and excellent chemical inertness to LMs under normal conditions,[33] SWCNTs can effectively separate two materials for a long time without being infiltrated by the LM. These advantages of the SWCNTs make them the most promising stretchable and reliable diffusion barrier to LMs.



Scheme 1. Schematics of the fabrication process and the morphology of the corrugated SWCNT/AgNP bilayer.

2.2. Barrier Performance of the SWCNT

To visualize the barrier performance of the SWCNTs, we investigated the cross-section of the LSCs between EGaIn and AgNP films by field-effect scanning electron microscopy (FE-SEM). Specimens with and without a spray-coated SWCNT barrier were prepared for comparison. To accelerate LMP, all of the samples were heated at 100 °C for 1 h in an oven before observation.[9] With the aid of the SWCNT barrier, the AgNP film under the barrier maintained its initial granular structure (**Figure 2**a) and no spreading front line around the EGaIn droplet was observed with the naked eye. The front line formation represents visible evidence of LM migration (Figures S1 and S2, Supporting Information).[9] In contrast, without the SWCNT barrier, a large amount of EGaIn intruded into the peripheral region. The AgNP film was swollen by massive mass transfer from the EGaIn droplet and peeled off from the underlying substrate (Figure 2b). Additional SEM observations and X-ray diffractometry (XRD)

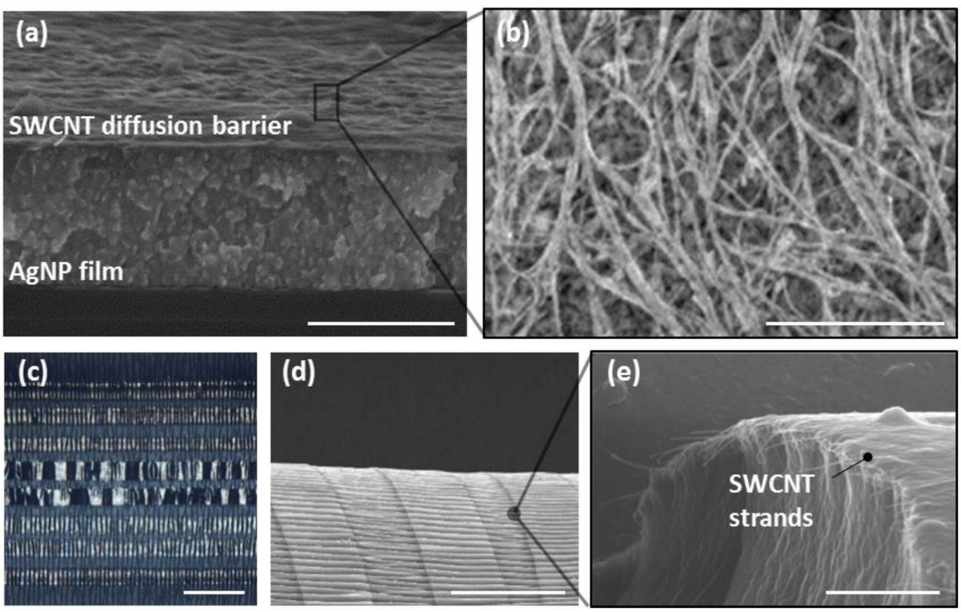


Figure 1. Morphology of the SWCNT/AgNP bilayer. a,b) SEM images of the SWCNT/AgNP bilayer. c–e) Morphology of the corrugated SWCNT/AgNP bilayer observed by optical microscopy and SEM. Scale bar: a) 1 µm, b) 1 µm, c) 500 µm, d) 200 µm, and e) 2 µm.

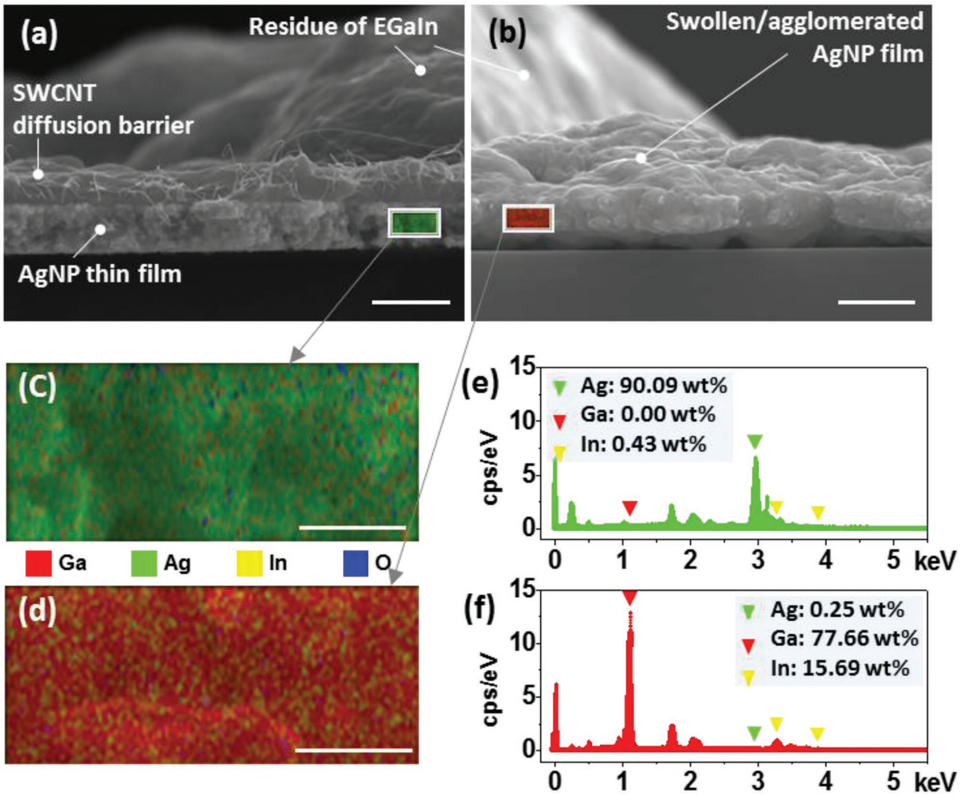


Figure 2. Verification of the diffusion blocking performance of the SWCNT layer on the AgNP film. a–d) Cross-sectional SEM images: a) AgNP film completely separated from EGaIn by the SWCNT layer. b) AgNP film impregnated with EGaIn. c) EDS images of the (a) AgNP film with the SWCNT barrier. d) EDS images of the (b) AgNP film with EGaIn penetration. e) EDS spectrum of the (c) AgNP film with the SWCNT barrier. f) EDS spectrum of the (d) AgNP film with EGaIn penetration. Scale bar: a) 1 µm, b) 1 µm, c) 250 nm, and d) 250 nm.

measurements in Figure S2 in the Supporting Information revealed that the infiltrated EGaIn chemically reacted with the AgNPs to form new chemical compositions,[47] aggregated the AgNP granules to a large cluster, and severely deformed the AgNP film (compare Figure S2b,d, Supporting Information). The energy dispersive X-ray spectroscopy (EDS) results corresponding to Figure 2a,b more clearly exhibit the effect of the SWCNT barrier over the AgNP film. The color mapping and spectrum results using EDS in accord with Figure 2a show that most of the AgNP film is in the initial state of Ag and almost no penetration of Ga or In occurred (Figure 2c,e). In contrast, the EDS results without the SWCNT barrier indicate that the most of the AgNP film was encroached by Ga and In and there is little observation of Ag peaks (Figure 2d,f). The reason that Ag was hardly detected at the AgNP film was possibly due to the limitation of the EDS. In measuring an elemental distribution of the sample, the electron beam of the EDS has limited penetration depth, which has difficulty in the quantitative analysis of 3D element distribution. In particular, the AgNPs in Figure 2b appears to be mostly covered with EGaIn penetrating into the AgNP thin film, which can block the detection of Ag and interfere exact quantitative analysis.

Next, to verify the chemical inertness of the SWCNTs to EGaIn, an SWCNT film was spray-coated onto an Si(100) wafer and kept in contact with the LM for 7 d. To speed up the chemical reaction, the temperature of the sample was retained at 100 °C in an oven. The EGaIn droplet was removed with a syringe without any chemical treatment to prevent the SWCNT from being chemically denatured by other factors. A Raman spectrum of the SWCNT layer was measured with a 532 nm laser source at locations where no residue of LM oxide was found. Figure S3a in the Supporting Information presents representative Raman spectra of SWCNTs exposed to EGaIn (blue) and a reference sample (black). The radial breathing mode peaks between 75 and 300 cm^{-1} in the inset, and the characteristic peaks at the D, G, and G′ positions denote the presence of SWCNTs. Notably, the *D* band (1330–1360 cm^{-1}) is used to evaluate the quality of the SWCNTs because this vibrational mode is associated with the disorder-induced band of the sp^2 carbon structure.[34] For this reason, a comparison of the intensities of the *D* and *G* bands, which is related to the tangential stretching mode of C–C, is an effective method for evaluating defects in the SWCNTs.[35,36] The intensity ratio of the D peak to the G peak of the SWCNTs in contact with EGaIn (I_D/I_G) is almost the same as that of the reference sample (≈7%). The inherent *D* band in our reference samples is thought to be –COOH functional groups, inherent structural defects, and carbonaceous impurities of the SWCNTs. The similar I_D/I_G ratios for both samples suggests that Ga and In do not change the sp^2 bonds of the SWCNTs to the sp^3 bonds and do not significantly increase the number of defects in the SWCNTs, unlike some reactive metal surfaces.[37,38]

Because of its impermeability and chemical inertness, the SWCNT is a good candidate for a diffusion barrier between EGaIn and AgNP. Furthermore, the SWCNT barrier is also applicable to other thin-film metals such as aluminum (Al), which is very vulnerable to Ga. For a demonstration, we dropped the EGaIn droplets on a bare Al foil and an SWCNT/Al foil. The SWCNT/Al foil was prepared by directly spraying the SWCNT suspension onto the Al foil. Figure S3b in the Supporting Information shows that the EGaIn droplet does not affect the Al foil under the SWCNT barrier, whereas the bare Al foil suffers from Ga-induced damage.

2.3. Electrical Characterization of the SWCNT/AgNP Bilayer

2.3.1. Measurement of the Contact Resistance

Several experiments were performed to confirm the barrier performance and characterize both the electrical properties and mechanical robustness of the LSC with an SWCNT/AgNP bilayer. Before the characterization, we first observed the cross-section of the spray-coated SWCNT to determine the thickness tendency of the SWCNT barrier with the number of coatings. The thickness of the film was measured by SEM at more than 10 locations, which were randomly chosen, and the results are presented in Figure S4 in the Supporting Information. The thickness of the film increased proportionally to the number of coatings and could be stacked as many as desired, ranging from tens of nanometers to hundreds of nanometers.

To measure the contact resistance of the SWCNT/AgNP contact with EGaIn, the transmission line method (TLM) was used.[39–41] The resistance measurements were performed by a Keithley 2400 SourceMeter SMU instrument (Tektronix, Beaverton, OR, USA). AgNP conducting lines with widths (Ws) of 2 mm, on which the LSCs with local SWCNT barriers were formed at different intervals (L = 2, 8, 16, and 32 mm, from the first LSC), were prepared by means of inkjet printing of the AgNPs. The area of the LSC was fixed to 2 mm × 2 mm, and the SWCNT barriers were patterned at the LSC locations using a mask before contact. The total resistance (R_T) from an EGaIn contact to another EGaIn contact at a distance of L can be expressed as

$$R_T = 2R_{EGaIn} + 2R_{SWCNT} + 2R_c + R_{AgNP} \tag{1}$$

where R_{EGaIn}, R_{SWCNT}, and R_{AgNP} are the resistance values of EGaIn, SWCNTs, and AgNPs, respectively, and R_c is the total contact resistance of the EGaIn/SWCNT or SWCNT/AgNP contact (**Figure 3**a). Assuming that the resistance inside EGaIn can be ignored and the resistance of the AgNP film is uniform over the entire section, Equation (1) can be approximated by

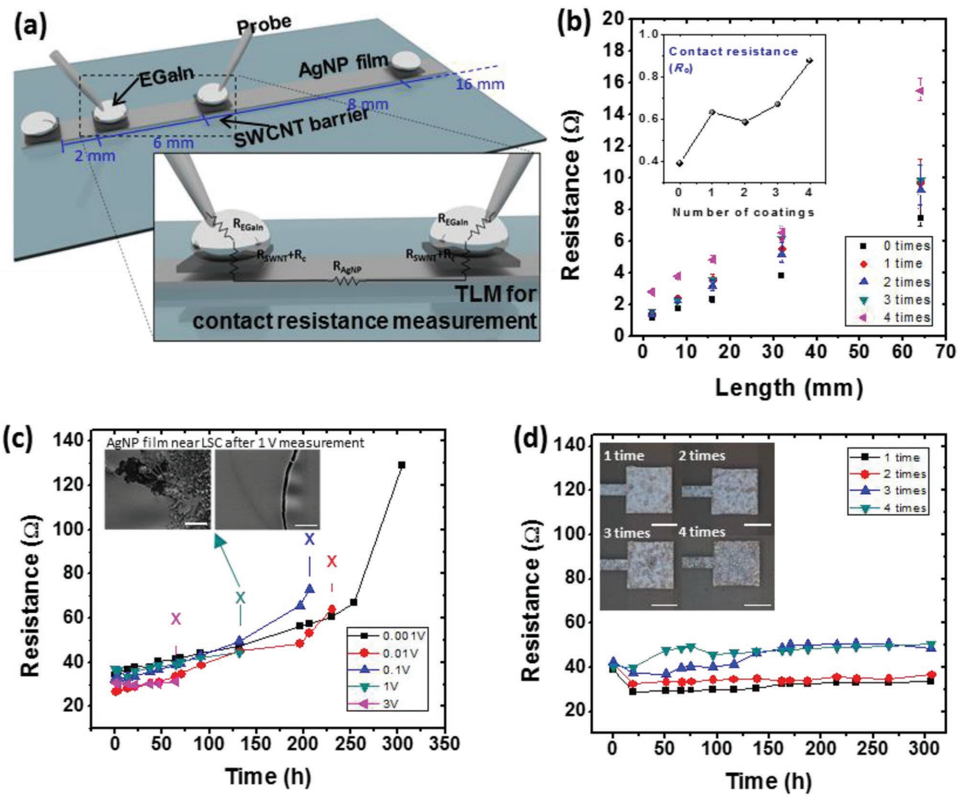


Figure 3. Electrical characterization of the SWCNT/AgNP bilayer. a) A schematic diagram of the TLM method for contact resistance measurements. R_{EGaIn}, R_{SWCNT}, and R_{AgNP} represent the resistance values of the EGaIn, SWCNTs, and AgNPs, respectively. R_c is the total contact resistance of the EGaIn/SWCNT or SWCNT/AgNP contact. b) Effect of the SWCNT spray-coating number on the contact resistance. TLM analysis was performed to measure the contact resistance. c) Dependence of the contact breakdown time on the measuring voltage and the time-dependent resistance change in AgNP directly exposed to LM. d) Resistance change in SWCNT/AgNP and EGaIn contact with time. The measuring voltage was 1 V. Scale bar: c) 10 µm.

$$R_T = 2(R_{SWCNT} + R_c) + R_s \frac{L}{W} \tag{2}$$

where R_s is the sheet resistance of the AgNP film. R_T was measured for each of the different L values, and the results are plotted against L in Figure 3b. According to Equation (2), the y intercepts in Figure 3b represent the sums of the total resistance values resulting from the SWCNT barriers and EGaIn contacts multiplied by 2. For the SWCNT coated zero to four times, the sums of the total contact resistances (R_{SWCNT} $+R_c$) are presented in Figure 3b. In the inset of Figure 3b, even without an SWCNT layer, the R_c at the LSC is ≈0.4 Ω due to both natural insulating oxide layer formation on the EGaIn surface[42] and the reduced effective contact area from the surface roughness and dewetting property at the contact.[43] The total contact resistance only slightly increases as the number of coating layers increases (0.6–0.9 Ω for 1–4 SWCNT coatings), indicating that the increase in R_{SWCNT} owing to the longer percolation path with the coating time is not very large.

2.3.2. Time-Dependent Resistance Changes of the LSCs

Measurements of the resistance variation over time were conducted at the LSC to determine the effect of degradation by the LM. Figure 3c shows the resistance changes of the LSCs without the SWCNT barrier measured with various voltages over 300 h. The LSCs were constructed on dogbone-shaped AgNP films on PDMS substrates, and EGaIn droplets were placed on both ends. In all samples, the resistance consistently increases with time. This result agrees with previous studies on several metals in contact with LMs.[21] Furthermore, for all of the samples, abrupt electrical breakdown rapidly occurs as the measuring voltage increases. This result indicates that the amount of current flow can also affect the breakdown of the LSCs. In Figure S5 in the Supporting Information, SEM and EDS images of the degraded LSC after the 300 h measurement at a measured voltage of 1 V are presented. In the EDS results in Figure S5c in the Supporting Information, large quantities of Ga and In are detected 250 µm away from the LSC. As a result, severe cracking and corrosion of the AgNP film are observed in Figure S5a,b in the Supporting Information, which could lower the conductivity of the AgNP film. In addition, additional oxide/IMCs formation of the migrated LM in the middle of the current path and the electrical breakdown of the Ag films weakened by EGaIn have also been suggested to be the probable causes.[15,21]

Figure 3d shows the electrical stability of the LSCs with SWCNT barriers. To determine the optimal amount of the coated layer, the SWCNT suspension was spray-coated onto the preprinted AgNP dogbone-shaped film on PDMS from one to four times. The voltage for measuring the resistance change was fixed at 1 V, which is considered to be sufficiently large to observe a change in the resistance with time. There is little change in the resistance of the SWCNT/AgNP LSCs with the only a slight change in the resistance during the measurement for each sample after ≈300 h. No sudden breakdown of the LSCs occurred during the 300 h measurements in all samples. This finding satisfactorily demonstrates the barrier effect of the SWCNT layer between the EGaIn and the AgNP thin film.

The thickness of the SWCNT barrier should be as thin as possible considering the fabrication process and contact resistance, according to TLM results in our experiments above. However, in an additional measurement after 1747 h, a slightly increased resistance was observed in the thinly coated samples, namely, from 38.7 to 42.3 Ω in the single coating and 41.6 to 56.5 Ω in the double coating. An increase in the resistance was not observed in the thicker barrier. This finding was considered a result of the penetration of the LM at some thinly coated locations due to the non-uniformity of the film thickness caused by the manual spray-coating process. This result also suggests that optimization of the amount of spray-coated SWCNTs to obtain a balance between the stable barrier formation and negligible contact resistance increase is required for practical applications. In the subsequent experiments, a double-coated SWCNT was selected as a diffusion barrier to ensure low contact resistance and adequate operating time.

2.3.3. Resistance Changes in the Corrugated LSCs under Stretching

To evaluate the potential of using an SWCNT/AgNP bilayer in a stretchable LSC, SWCNT layer was formed on a corrugated AgNP stretchable conductor and the mechanical reliability of the LSC with a corrugated SWCNT/AgNP bilayer was investigated. The corrugated SWCNT/AgNP LSCs was fabricated on a pre-stretched PDMS substrate in the same manner as our previous studies.[25–27] The bilayer showed a well-formed corrugated structure similar to a pristine AgNP film because of the very small thickness and entangled structure of the SWCNT layer, as shown in Figure 1. This accordion bellow-like structure endows structural redundancy through the folded state to minimize the mechanical stress in the film, even under tensile strain conditions. Repeated tensile tests of both LSCs with the corrugated SWCNT/AgNP bilayer and the corrugated AgNP monolayer were carried out to compare their mechanical reliability and robustness (**Figure 4**a). Before the test, each sample was left in the oven at 100 °C for 2 h to be sufficiently damaged by the LM. During the 3000 cycles of the test (ε = 30%), only a small resistance change due to the fluidity of the EGaIn was observed in the case of the corrugated SWCNT/AgNP LSC (graphs with the blue line in Figure 4a). In contrast, the AgNP only LSC tested under the same conditions showed a steady increase in resistance during the repeated tensile test; finally, a 1.7 times increase in resistance occurred after 3000 cycles (graphs with the red line in Figure 4a). To determine that the major reason of the resistance increase is LM penetration, the bottom sides of both LSCs were analyzed by an optical microscope through the transparent PDMS substrate (Figure 4b). Without the SWCNT, dark spots were observed over the entire area of the AgNP film under EGaIn just after 2 h under 100 °C. These dark spots are due to the penetration of EGaIn. After repeated stretching tests, the bottom of the LSCs was observed again after 140 h. In Figure 4c, the dark spots under the EGaIn without the SWCNT were widened toward other regions. Even worse, some of the regions inside the dark spots appeared to be easily damaged by mechanical stress, resulting in cracks along the orthogonal direction of the wrinkles during the stretching test. According to our previous

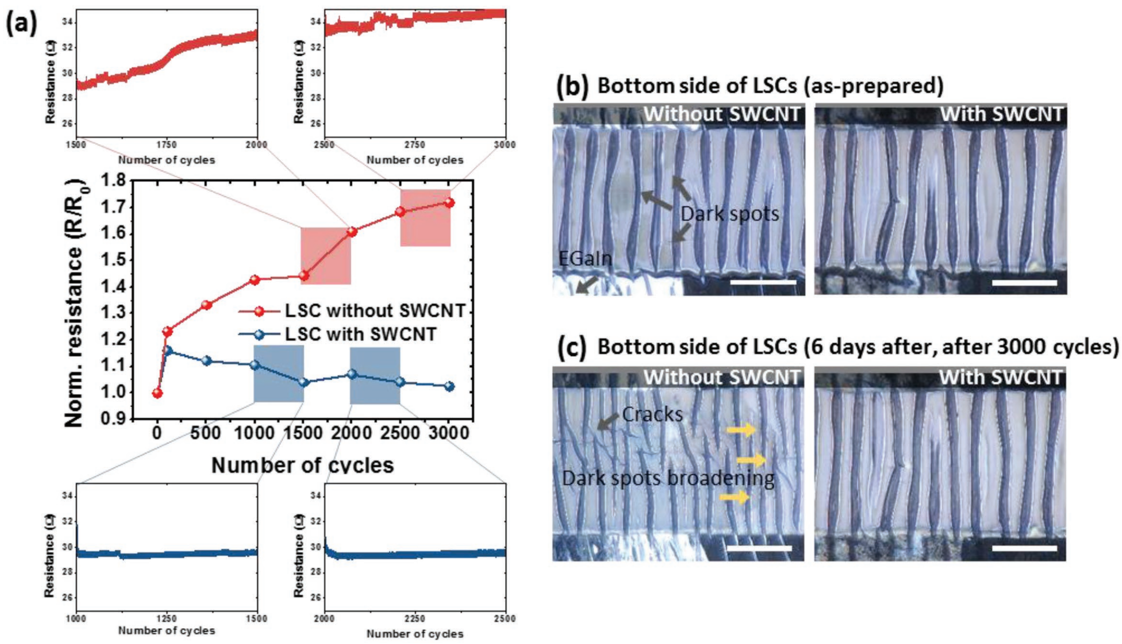


Figure 4. Mechanical stability of the SWCNT/AgNP LSCs. a) Resistance change in the SWCNT/AgNP film in contact with EGaIn during repetitive tensile tests 140 h after fabrication. In all of the stretching cycles, the applied strain was 30%, and the stretching speed was 500 mm min^{-1}. b,c) Corrugated LSCs with and without SWNCTs: b) as-prepared. c) 6 d after, after 3000 repeated tensile tests.

experimental results, LMP seems to induce the peeling and cracking of the AgNP film, which degrades the mechanical robustness of the denatured film. It is believed that when the stress due to the stretching and bending deformation is continuously delivered to the weakened AgNP film, small defects accumulate in the film and increase the resistance of the LSC.

2.4. LM-Based Chip Bonding with the Corrugated SWCNT/AgNP Bilayer

With a minimal increase in the contact resistance and no damage on the stretchable conductors demonstrated above, an LM-based chip bonding technique for SHE system integration was developed. In general, most studies of the application of surface mount device (SMD)-type chips have used a rigid solder (e.g., epoxy-based solder) for their electrical connections. However, in most SHE systems, the transition region between the solder and soft substrates or conductors can be easily damaged under deformation conditions because of the difference in their elastic moduli. We have reported a rigid island approach to protect such a transition region from the external stress.[25] Nevertheless, stress can occur when the force induced by friction or mechanical contact is directly applied onto the SMD. In our LM-based bonding approach, by replacing the rigid solder with an LM, the rigid chips and the soft substrate or conductors are mechanically decoupled, alleviating the stress that concentrates at the interface between the two materials with considerably different elastic moduli.

A schematic diagram of the LM bonding strategy is shown in **Figure 5**a, b. EGaIn droplets, which act as the solder materials, are dispensed onto the soft circuit pads. To ensure the long-term reliability of the LM contact, an SWCNT barrier is spray-coated onto the soft circuit pads. For robust chip bonding and EGaIn confinement, pure epoxy bank structures are also dispensed between the pads. These pure epoxy banks effectively separate the pads to prevent an electrical short between the LM, which can occur during the chip placement process and circuit operation. The epoxy structure can also act as a rigid island, preventing large deformation around the chip area (Figure S6, the complete manufacturing process is described in the Supporting Information).

To investigate the robustness of the LM soldering technique to external mechanical stress, tensile and compressive stresses were applied to the stretchable substrate and soldered chips, respectively. For the characterization, a 0 Ω jumper resistor (1/8 W; Yageo, Taipei, Taiwan) was used.[44] The resistance change in the jumper resistors and the corrugated SWCNT/AgNP bilayer electrodes soldered by EGaIn were measured during stretching (Figure 5c). The resistance was plotted on the graph during one cycle of stretching-releasing, depending on the strain applied to the substrate. There was a 4% change in the resistance compared to the initial value in one cycle, probably due to the LM movement along the deformation of the substrate. This change was increased to ≈10% during 500 times cycles, suggesting an additional improvement such as wetting properties between the SWCNT barrier and EGaIn or encapsulation of the entire device. The impressive advantage of LM soldering is the reliability at normal pressures, as shown in Figure 5d. For comparison, LED chips soldered on corrugated AgNP electrodes by rigid epoxy and EGaIn were prepared on stretchable substrates. The soldered chips were placed under a slowly increasing vertical pressure up to 500 gf. Unlike the LM-soldered LED, a sudden increase in the resistance during the vertical pressure occurred to the epoxy-soldered LED. On the other hand, the resistance of LEDs using LM bonding did

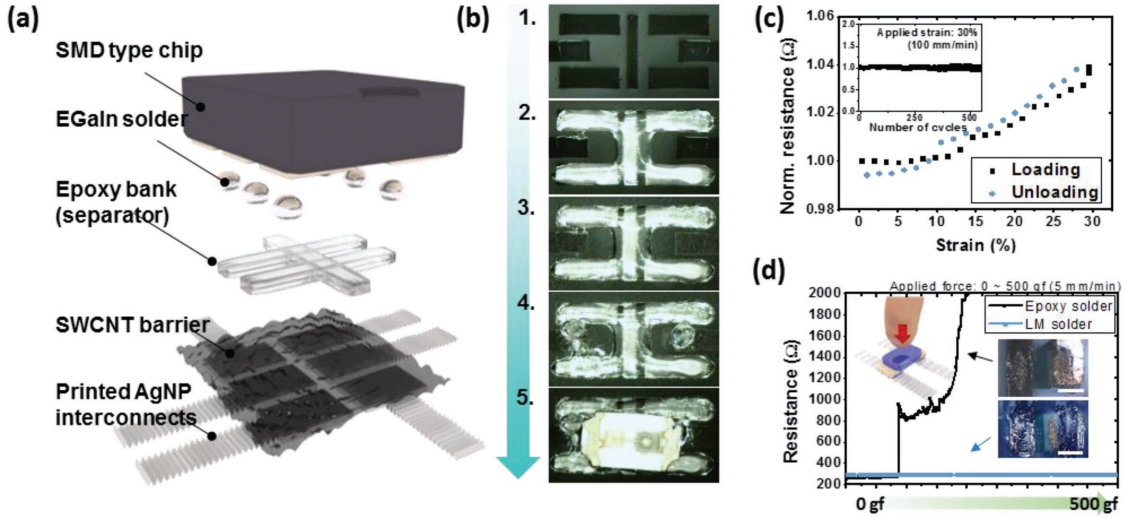


Figure 5. LM-based chip bonding technique for soldering a rigid chip on a stretchable circuit using an impermeable corrugated SWCNT/AgNP bilayer. a) Schematic of LM bonding. b) Fabrication process flow: 1) printing of AgNPs on pre-stretched PDMS. 2) Dispensing the epoxy bank structure. 3) Spray coating the SWCNT layer. 4) Dispensing the EGaIn solder. 5) Picking and placing the LED chips. c) Resistance change in the soldered 0 Ω resistor chip with LM bonding. The inset graph shows the resistance change under repeated stretching cycles at a speed of 100 mm min^{-1}. d) Comparison of the resistance change in an epoxy-soldered LED chip and an LM-soldered LED chip. For ≈18 s, a normal force was applied directly to the chips and gradually increased from 0 to 500 gf. The pressing speed was 5 mm min^{-1}.

not change with the pressure, and the mechanical stability was very satisfactorily maintained over 200 repetitive pressure cycles (Figure S7, Supporting Information). As displayed in the inset images of Figure 5d, the AgNP film under the epoxy did not exhibit a corrugated structure after the pre-stretching process. This result implies that the interface between the AgNP thin films and solders were vulnerable to external stress. On the other hand, the thin film under the LM was not mechanically adhered to the solder, resulting in spontaneous wrinkles during the fabrication process. Additionally, our method is much more robust against mechanical stress since the force applied to the chip is scarcely transferred to the underlying AgNP film. This result ensures that the LM soldering technique has the potential to be utilized in soft devices such as wearable mobile healthcare devices and stretchable interactive displays that are generally exposed to various kinds of mechanical stress.

Based on the optimized SWCNT/AgNP LSCs and LM-based chip bonding technique, we have further demonstrated a seven-segment LED display containing two LEDs in each segment pixel (**Figure 6**a,b). The proposed bonding process is completely compatible with the inkjet-printed stretchable circuit fabrication process introduced in our previous studies.[25–27] Consequently, the process is suitable for complicated and sophisticated circuit configurations consisting of microscale chips and conducting elements. Through the fabrication process described above, the SMD-type LED chips were robustly integrated with the stretchable circuit fabricated on a soft PDMS substrate. The performance of the stretchable seven-segment display was monitored for 10 d of display operation. Figure 6c shows that there is no noticeable reduction in the luminance of the LEDs of the different numbers after operation for 10 d, again confirming the improved lifetime of the LSCs. The seven-segment circuit operated well, even under stretching conditions owing to the corrugated structure of the SWCNT/AgNP bilayer (Figure 6d). The stretchable seven-segment circuit could be stretched with a strain of 30% along the one axis and even could be stretched along the two perpendicular axes with strain of 20% in each axis (a total area increase of 144%). Notably, the display could not be stretched beyond the degree of the initial pre-stretching used during the fabrication process. Within the range of the pre-stretched strain, the display was able to withstand more than a hundred stretching–releasing cycles (Figure S8, Supporting Information). Furthermore, the proposed methodology was capable of being utilized in an SMD chip with two or more multiple contact pads. A three-color LED chip with four contact pads was bonded on the printed stretchable circuit by LM-based bonding technique. The current–voltage (I-V) characteristics of the three-color LED chip is shown in Figure 6e, and the results show a slight change in the current to voltage sweep during the 2D stretching of 20% in the two perpendicular directions.

3. Conclusion

In conclusion, we have reported a corrugated SWCNT diffusion barrier on an AgNP thin film for reliable and stretchable LSCs. We clearly established that Ga-based LMs degrade metallic thin films including nanoparticle/nanowire-based metallic thin films when they are in contact with each other. Consequently, without solving this critical technical issue, LMs have limited usage in practical stretchable electronic circuits. We used densely packed SWCNTs to effectively block LM interdiffusion, thus enabling LSCs with much longer lifetimes. In addition, the ultrathinness and mechanical robustness of the SWCNT layer rendered the barrier-induced contact resistance negligibly small and produced a crack-free, long-term reliable, and stretchable corrugated bilayer structure. The resulting corrugated SWCNT/AgNP bilayer showed no significant change in the stretchability

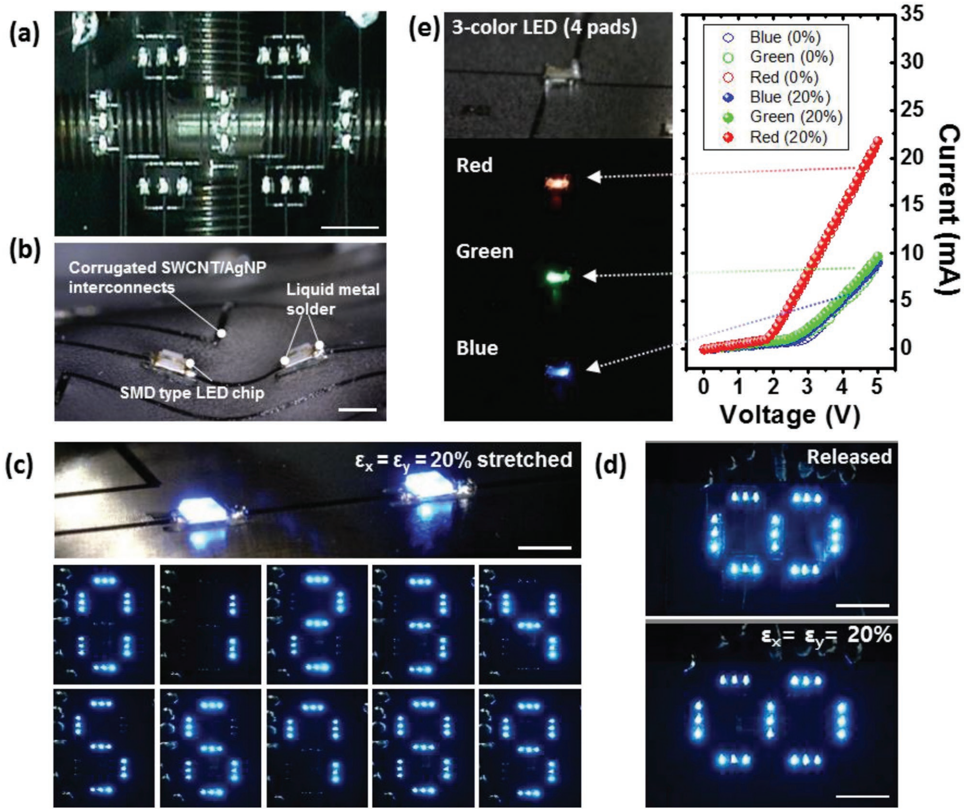


Figure 6. Demonstration of the seven-segment stretchable circuit with LM soldering and the corrugated SWCNT/AgNP bilayer. a,b) Optical images of the stretchable seven-segment circuit. c) Operation of the seven-segment circuit 10 d after fabrication. d) Operation of the seven-segment circuit during 2D stretching. e) *I–V* characteristics of the LM-soldered three-color LED chip with four contact pads.

and conductivity, even after exposure to EGaIn for several hundred hours. In addition, the long-term reliability of the bilayer enabled us to construct a stably operating stretchable seven-segment display and potential full-color display. In fact, the inhomogeneity of the SWCNT film from the lab-scale coating process and limitations in its much finer feature patterning may be potential obstacles for fabricating a more sophisticated stretchable system. However, it is expected these technical challenges can be overcome if other appropriate printing processes, such as inkjet[45] or electro-hydrodynamic[46] methods are adopted.

4. Experimental Section

Materials: AgNP ink (AgNP dispersion, DGP-40LT-15C; ANP., South Korea) was used to form the AgNP thin films. To prepare the SWCNT suspension, carboxylic acid-functionalized SWCNT powder (SWCNT-COOH, average length 5–30 µm, average diameter 1–2 nm; Fine Chemical Industry, South Korea) and sodium SDBS (Sigma–Aldrich, St. Louis, MO, USA) surfactant were dispersed in DI water. Eutectic gallium–indium (75.5% Ga and 24.5% In by weight; Sigma–Aldrich) was used to observe the effect of the LM on the AgNP thin film. PDMS (Sylgard 184; Dow Corning, Midland, MI, USA) was used as a stretchable substrate at a mixing ratio of 20:1 with a curing agent.

Preparation of the SWCNT Suspension: SWCNT powder was dispersed in DI water by tip sonication (VCX 130; Sonics & Materials, Newtown, CT, USA) for 80 min. SDBS was added to the dispersion as a surfactant to improve the dispersion of the SWCNTs in DI water, followed by an additional tip sonication process for 80 min. The mixing ratio of the SWCNT-COOH powder, SDBS surfactant, and DI water was 1:5:5000 by weight. The dispersion was filtered during the final process using quantitative filter paper (Grade 541; Whatman, Maidstone, UK) to minimize spray-nozzle clogging.

Preparation of the Substrate: An Si(100) wafer with thermal oxide or soda-lime glass was used as the rigid substrate to prepare the non-stretchable AgNP thin film for observation of the effect of EGaIn on the SWCNTs and AgNPs. The Si wafers and glasses were immersed in acetone and isopropyl alcohol for 1 h each with ultrasonication before AgNP coating. The stretchable substrate was prepared on a PDMS substrate. PDMS with a curing agent was poured and spin-coated onto the glasses at a spin speed of 300 rpm for 60 s after a sufficient degassing process. The PDMS substrate was then thermally cured at 80 °C for 2 h. To increase the number of hydrophilic functional groups on the PDMS surface, the surface of PDMS was treated with UV-O_3 cleaner (AH1700; Ahtech LTS, Anyang, South Korea) for 25 min. When imparting an additional corrugated structure and stretchability to the hybrid film, the PDMS substrate was pre-stretched before surface treatment. For the uniaxial stretching mode, the applied strain $\varepsilon_x = 50\%$. For the omnidirectional stretching mode, the applied strain $\varepsilon_x = \varepsilon_y = 30\%$.

Preparation of the SWCNT/AgNP Bilayer: To observe the general effect of the LM on the AgNP film and SWCNTs, Ag thin films were prepared on a rigid substrate by spin coating of the AgNP ink at 600 rpm for 60 s. For electrical characterization of the AgNPs and SWCNT/AgNP hybrid film, the AgNP ink was patterned in a dogbone shape on a PDMS substrate by an inkjet printing system (DMP-2831; Fujifilm Dimatix, Santa Clara, CA, USA). The pattern size was 0.2 mm × 10 mm for the resistance measurement in the static state and 0.2 mm × 15 mm for the stretchability evaluation. The spacing of the inkjet-printed droplets was set to 32 µm, and only one nozzle was used during the printing process. During the printing process, the substrate temperature was maintained at 60 °C. The samples were then sintered at 150 °C for 2 h to increase the connectivity between the AgNPs. For formation of the SWCNT barrier on the AgNP film, the SWCNT suspension was spray-coated by an airbrush. To uniformly cover the entire substrate, ≈30 cm of clearance between the sample and airbrush was maintained during the coating process, and the coating area was limited to 60 mm × 60 mm. The amount of the suspension used in each coating was 0.5 mL. During the spray-coating process, the substrate was heated at 150 °C to reduce damage to the AgNPs by water in the SWCNT suspension.

Field-Effect Scanning Electron Microscopy: The corrugated SWCNT/AgNP hybrid films were observed with an S-4800 SEM (Hitachi, Tokyo, Japan). The acceleration voltage was 20 kV for clear observation of the SWCNT strands. To analyze the elemental distributions and morphologies of the hybrid films, a JSM-7800F Prime EDS (JEOL, Tokyo, Japan) was used. All of the samples were kept in contact with EGaIn at 100 °C for 24 h prior to the measurements to promote the movement and reactivity of EGaIn.

Raman Spectroscopy: Raman spectroscopy was employed to study defects of the SWCNTs. LabRAM HV Evolution spectrometer (Horiba, Kyoto, Japan) using a 532 nm laser as the light source was used. The analyzed range was from 0 to 3000 cm^{-1}, which sufficiently covers the range of peaks generated by the SWCNT layer and contains the radial breathing mode (75–300 cm^{-1}). To confirm the effect of EGaIn on the SWCNTs, the SWCNT layer was coated onto an Si(100) wafer and then kept in contact with EGaIn at 100 °C for 0–7 d.

X-Ray Diffractometry: XRD was used to verify chemical denaturation of the AgNP film by EGaIn. D8-Advance diffractometer (Bruker Miller, Karlsruhe, Germany) with Cu $K\alpha_1$ radiation was utilized. The measurement circle diameter was 435 mm, and the voltage was 40 kV. The AgNPs were spin-coated on an Si(100) wafer with an oxide (600 rpm for 60 s). After sintering, an EGaIn droplet was placed onto the AgNP film and heated at 100 °C for 937 h. Prior to XRD analysis, EGaIn was drawn off with a syringe.

Strain Distribution Analysis: The digital image correlation method was used for strain distribution mapping around the epoxy structure. The fine particles were uniformly scattered on the surface to form random speckle patterns. The VIC-2D program (Correlated Solutions, Columbia, SC, USA) was used for tracking the movement of the speckle patterns. The strain distribution was calculated by comparing the positions of the meshes from the speckle patterns.

Repeated Tensile Test: Repeated tensile tests were performed with corrugated LSCs with and without the SWCNT barrier (15 mm × 0.2 mm). During the test cycles, the specimens were stretched by 30% of the initial length at a speed of 100–500 mm min^{-1}. The resistance was measured by a Keithley 2400 SourceMeter SMU instrument (Tektronix, Beaverton, OR, USA).

Repeated Pressure Test. Repeated pressure tests were performed to compare the mechanical and electrical robustness under direct normal pressure conditions. The pressure was measured by a DGT-1 digital push pull force gauge (DigiTech, Osaka, Japan), and the repeated pressure was loaded by an ASM-1000 test stand (DigiTech, Osaka, Japan).The speed of the force gauge was set to 5 mm min^{-1}. The resistance was measured with a Keithley 2400 SourceMeter SMU instrument (Tektronix).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was partly supported by the Institute for Information and Communications Technology Promotion (IITP) grant funded by the Korean government (MSIP No. 2017-0-00048, the Development of Core Technologies for Tactile Input/Output Panels in Skintronics (Skin Electronics)), and the Center for Advanced Soft-Electronics funded by the Ministry of Science, ICT and Future Planning as Global Frontier Project (CASE-2015M3A6A5065309).

Conflict of Interest

The authors declare no conflict of interest.

Keywords

carbon nanotubes, diffusion barrier, liquid metal, liquid metal–solid metal contact, stretchable electronics

Received: August 28, 2018
Revised: September 21, 2018
Published online: October 16, 2018

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